Fundamentals Physics

Twelfth Edition

Halliday

Chapter 4

Motion in Two and Three Dimensions

Section 4.1 Position and Displacement

Position and Displacement Learning Objectives

- Draw two-dimensional and three-dimensional position vectors for a particle, indicating the components along the axes of a coordinate system.
- On a coordinate system, determine the direction and magnitude of a particle's position vector from its components, and vice versa.
- Apply the relationship between a particle’s displacement vector and its initial and final position vectors.

Position Vector: Definition

- A **position vector** locates a particle in space
  - Extends from a reference point (origin) to the particle

$$\vec{r} = x\hat{\mathrm{i}} + y\hat{\mathrm{j}} + z\hat{\mathrm{k}}$$ **Equation (4.1.1)**

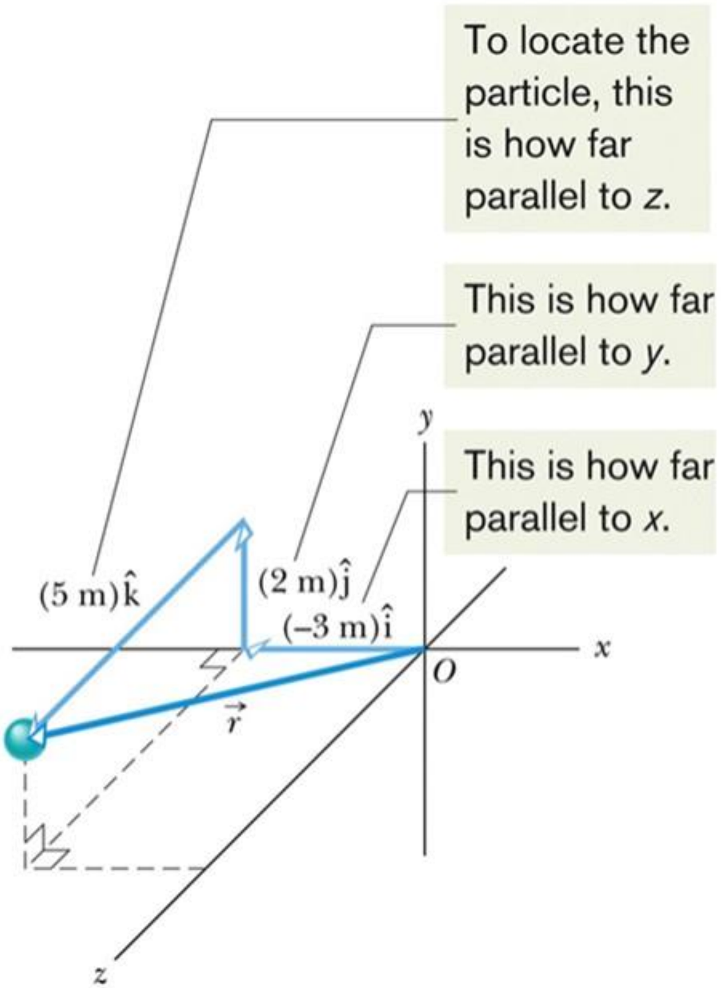
Example

Position vector (−3m, 2m, 5m)

$$\vec{r} = (-3\ \mathrm{m})\hat{\mathrm{i}} + (2\,\mathrm{m})\hat{\mathrm{j}} + (5\,\mathrm{m})\hat{\mathrm{k}}$$

Position Vector = Sum of Components

Figure 4.1.1



Position vs. Displacement

- Change in position vector is a **displacement**

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1. \qquad \textbf{Equation (4.1.2)}$$

- We can rewrite this as:

$$\Delta\vec{r} = (x_2 - x_1)\hat{\mathrm{i}} + (y_2 - y_1)\hat{\mathrm{j}} + (z_2 - z_1)\hat{\mathrm{k}} \qquad \textbf{Equation (4.1.3)}$$

- Or express it in terms of changes in each coordinate:

$$\Delta\vec{r} = \Delta x\hat{\mathrm{i}} + \Delta y\hat{\mathrm{j}} + \Delta z\hat{\mathrm{k}} \qquad \textbf{Equation (4.1.4)}$$

Section 4.2 Average and Instantaneous Velocity

Average Velocity and Instantaneous Velocity Learning Objectives

- Identify that velocity is a vector quantity and thus has both magnitude and direction and also has components.
- Draw two-dimensional and three-dimensional velocity vectors for a particle, indicating the components along the axes of the coordinate system.
- In magnitude-angle and unit-vector notations, relate a particle's initial and final position vectors, the time interval between those positions, and the particle's average velocity vector.
- Given a particle's position vector as a function of time, determine its (instantaneous) velocity vector.

Average Velocity: Definition

- **Average velocity** is
  - A displacement divided by its time interval

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{\Delta t}$$ **Equation (4.2.1)**

- We can write this in component form:

$$\vec{v}_{\text{avg}} = \frac{\Delta x\hat{\text{i}} + \Delta y\hat{\text{j}} + \Delta z\hat{\text{k}}}{\Delta t} = \frac{\Delta x}{\Delta t}\hat{\text{i}} + \frac{\Delta y}{\Delta t}\hat{\text{j}} + \frac{\Delta z}{\Delta t}\hat{\text{k}}$$ **Equation (4.2.2)**

Average Velocity: Example

A particle moves through displacement (12 m)i + (3.0 m)k in 2.0 s:

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{\Delta t} = \frac{(12\,\text{m})\hat{\text{i}} + (3.0\ \text{m})\hat{\text{k}}}{2.0\,\text{s}} = (6.0\ \text{m/s})\hat{\text{i}} + (1.5\ \text{m/s})\hat{\text{k}}$$

Average Velocity vs. Instantaneous Velocity

Instantaneous velocity is defined as:

- The velocity of a particle at a single point in time
- The limit of avg. velocity as the time interval shrinks to 0

$$\vec{v} = \frac{d\,\vec{r}}{dt}$$ **Equation (4.2.3)**

Graphical Average vs. Instantaneous Velocity

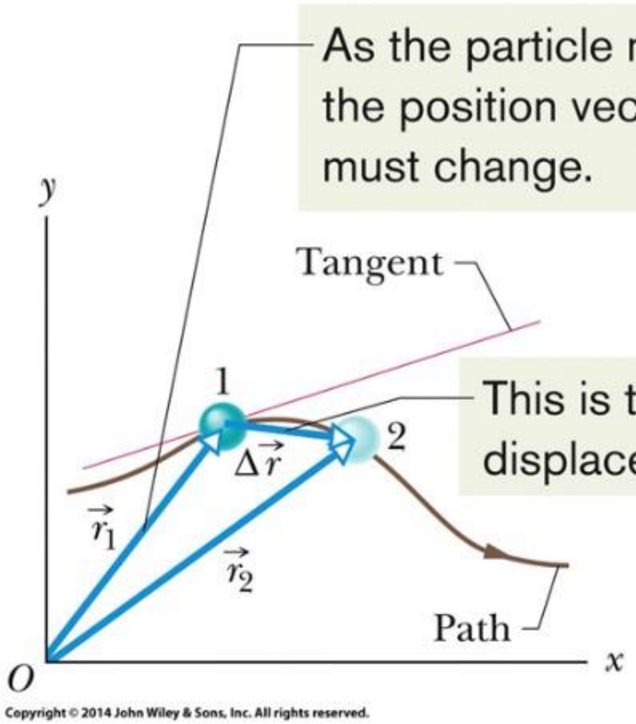


Figure 4.2.1

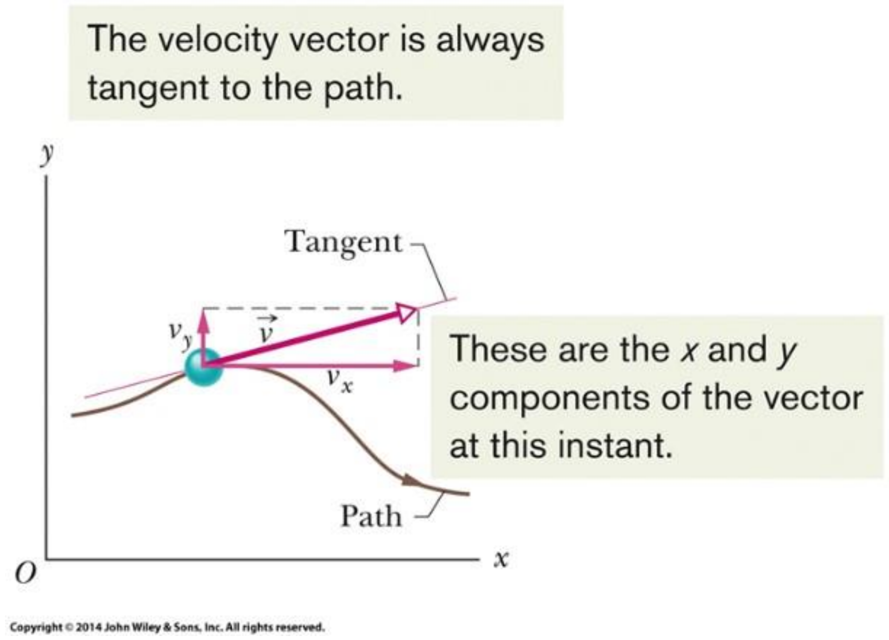


Figure 4.2.2

Instantaneous Velocity: Vector Notation

The direction of the instantaneous velocity $\vec{v}$ of a particle is always tangent to the particle's path at the particle's position.

- In unit-vector form, we write:

$$\vec{v} = \frac{d}{dt}\left(x\hat{\mathrm{i}} + y\hat{\mathrm{j}} + z\hat{\mathrm{k}}\right) = \frac{dx}{dt}\hat{\mathrm{i}} + \frac{dy}{dt}\hat{\mathrm{j}} + \frac{dz}{dt}\hat{\mathrm{k}}$$

- Which can also be written:

$$\vec{v} = v_x\hat{\mathrm{i}} + v_y\hat{\mathrm{j}} + v_z\hat{\mathrm{k}}$$ **Equation (4.2.4)**

Magnitudes of Instantaneous Velocity Vector Components

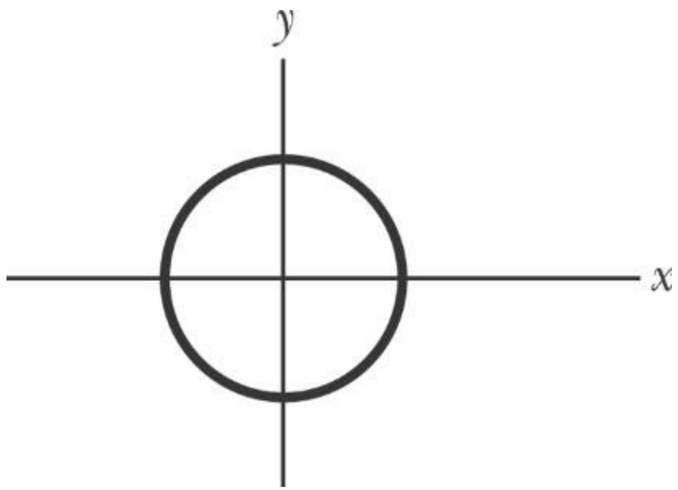
$$v_x = \frac{dx}{dt}, \quad v_y = \frac{dy}{dt}, \quad \text{and} \quad v_z = \frac{dz}{dt}. \qquad \textbf{Equation (4.2.5)}$$

Subscripts *x*, *y*, and *z* denote the vector direction

- Note: a velocity vector does not extend from one point to another, only shows direction and magnitude

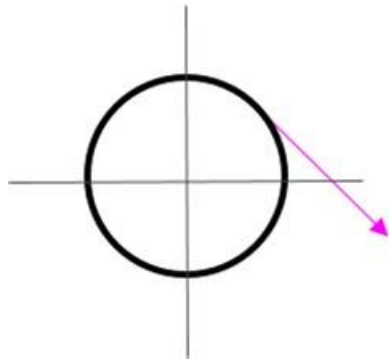
Instantaneous Velocity Checkpoint #1 Question

The figure shows a circular path taken by a particle. If the instantaneous velocity of the particle is $\vec{v} = (2\,\text{m/s})\hat{\text{i}} - (2\text{ m/s})\hat{\text{j}}$, through which quadrant is the particle moving at that instant if it is traveling (a) clockwise and (b) counterclockwise around the circle? For both cases, draw $\vec{v}$ on the figure.

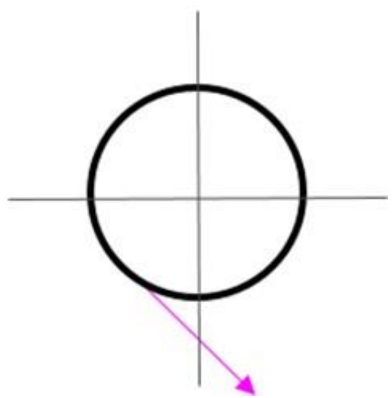


Instantaneous Velocity Checkpoint #1 Answer

(a) Quadrant I



(b) Quadrant III



Section 4.3 Average and Instantaneous Acceleration

Average Acceleration and Instantaneous Acceleration Learning Objectives

- Recognize that acceleration is a vector quantity, and thus has both magnitude and direction.
- Draw two-dimensional and three-dimensional acceleration vectors for a particle, indicating the components.
- Given the initial and final velocity vectors of a particle and the time interval, determine the average acceleration vector.
- Given a particle's velocity vector as a function of time, determine its (instantaneous) acceleration vector.
- For each dimension of motion, apply the constant-acceleration equations (Chapter 2) to relate acceleration, velocity, position, and time.

Average vs. Instantaneous Acceleration: Definitions

- **Average acceleration** is
  - A change in velocity divided by its time interval

$$\vec{a}_{avg} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t} = \frac{\Delta\vec{v}}{\Delta t}$$ **Equation (4.3.1)**

$$v_2 = \text{final velocity},\ v_1 = \text{initial velocity}$$

- **Instantaneous acceleration** is again the limit $t \to 0$:

$$\vec{a} = \frac{d\vec{v}}{dt}$$ **Equation (4.3.2)**

Instantaneous Acceleration: Vector Notation

- We can write Eq. 4.3.2 in unit-vector form:

$$\vec{a} = \frac{d}{dt}\left(v_x\hat{\mathrm{i}} + v_y\hat{\mathrm{j}} + v_z\hat{\mathrm{k}}\right)$$

$$= \frac{dv_x}{dt}\hat{\mathrm{i}} + \frac{dv_y}{dt}\hat{\mathrm{j}} + \frac{dv_z}{dt}\hat{\mathrm{k}}$$

Magnitude of Instantaneous Acceleration Components

- We can rewrite as:

$$\vec{a} = a_x\hat{\text{i}} + a_y\hat{\text{j}} + a_z\hat{\text{k}} \qquad \textbf{Equation (4.3.3)}$$

$$a_x = \frac{dv_x}{dt}, \quad a_y = \frac{dv_y}{dt}, \quad \text{and} \quad a_z = \frac{dv_z}{dt} \qquad \textbf{Equation (4.3.4)}$$

- To get the components of acceleration, we differentiate the components of velocity with respect to time

Instantaneous Acceleration Vector Acting on a Particle Traveling on a Path

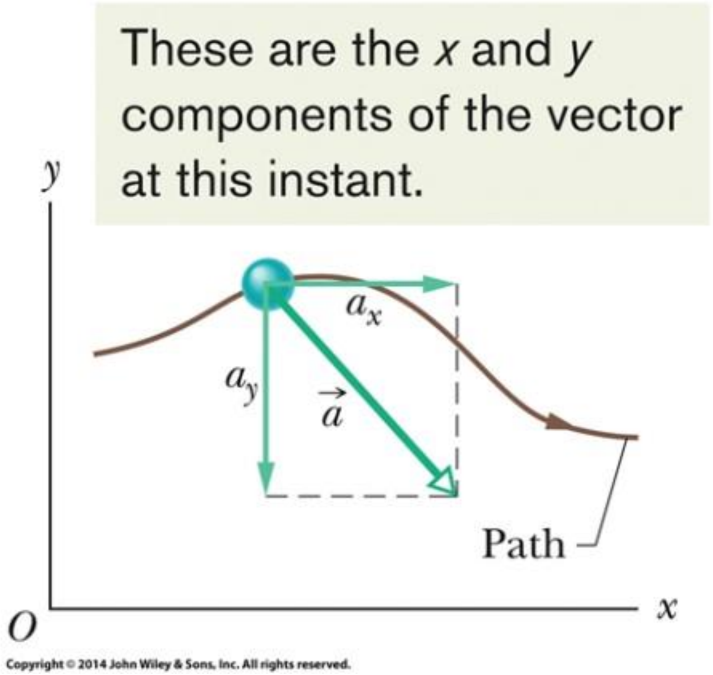


Figure (4.3.1)

Acceleration Vector Specific to Particular Points

- Note: as with velocity, an acceleration vector does not extend from one point to another, only shows direction and magnitude

Average vs. Instantaneous Acceleration: Checkpoint #2 Question

Below are four descriptions of the position (in meters) of a puck as it moves in an xy plane:

$(1)\ x = -3t^2 + 4t - 2 \quad \text{and} \quad y = 6t^2 - 4t$

$(2)\ x = -3t^3 - 4t \quad \text{and} \quad y = -5t^2 + 6$

$(3)\ \vec{r} = 2t^2\hat{\mathrm{i}} - (4t + 3)\hat{\mathrm{j}}$

$(4)\ \vec{r} = \left(4t^3 - 2t\right)\hat{\mathrm{i}} + 3\hat{\mathrm{j}}$

Are the x and y acceleration components constant? Is the acceleration vector constant?

Average vs. Instantaneous Acceleration: Checkpoint #2 Answers

1) x : yes, y : yes, a : yes
2) x : no, y : yes, a : no
3) x : yes, y : yes, a : yes
4) x : no, y : yes, a : no

Section 4.4 Projectile Motion

Projectile Motion Learning Objectives

- On a sketch of the path taken in projectile motion, explain the magnitudes and directions of the velocity and acceleration components during the flight.
- Given the launch velocity in either magnitude-angle or unit-vector notation, calculate the particle's position, displacement, and velocity at a given instant during the flight.
- Given data for an instant during the flight, calculate the launch velocity.

Projectile: Definition

- A **projectile** is
  - A particle moving in the vertical plane
  - With some initial velocity
  - Whose acceleration is always free-fall acceleration (g)
- The motion of a projectile is **projectile motion**
- Launched with an initial velocity $\vec{v}_0$

$$\vec{v}_0 = v_{0x}\hat{\mathrm{i}} + v_{0y}\hat{\mathrm{j}}$$ **Equation (4.4.1)**

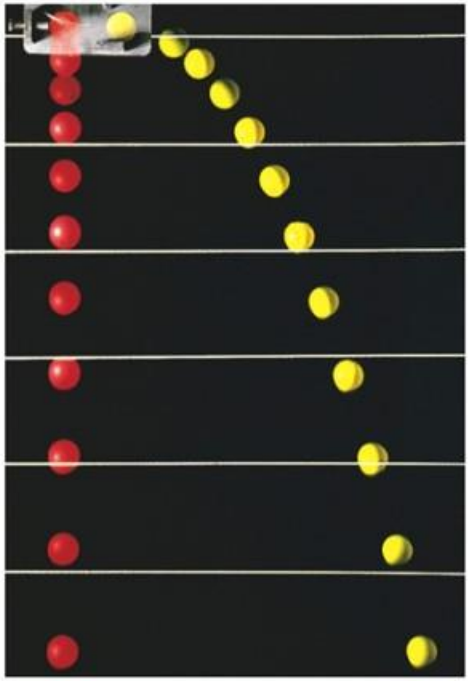
$$v_{0x} = v_0 \cos\theta_0 \quad \text{and} \quad v_{0y} = v_0 \sin\theta_0$$ **Equation (4.4.2)**

In projectile motion, the horizontal motion and the vertical motion are independent of each other; that is, neither motion affects the other.

Effect of Horizontal Velocity Component

The red and yellow balls are dropped at the same time, but the yellow ball was shot horizontally (to the right) at the start of the fall.

- vertical motion the same for red vs. yellow balls
- horizontal motion different
- red and yellow balls strike the ground at the same time since both have the same (zero) initial velocity in the vertical direction.



Richard Megna/Fundamental Photographs

Figure 4.4.3

Projectile Motion Checkpoint #3

At a certain instant, a fly ball has velocity $\vec{v} = 25\hat{\mathrm{i}} - 4.9\hat{\mathrm{j}}$

(the x axis is horizontal, the y axis is upward, and v is in meters per second). Has the ball passed its highest point?

Answer:

Yes. The y-velocity is negative, so the ball is now falling.

Horizontal Component of Projectile Motion

Horizontal motion: No acceleration, so velocity is constant (recall Eq. 2.4.5):

$$x - x_0 = v_{0x} t.$$

$$x - x_0 = \left(v_0 \cos\theta_0\right) t. \quad \textbf{Equation (4.4.3)}$$

Vertical Component of Projectile Motion

Vertical motion: Acceleration is always -g (recall Eqs. 2.4.5, 2.4.1, 2.4.6)

$$y - y_0 = v_{0y}t - \frac{1}{2}gt^2$$

$$= (v_0 \sin\theta_0)t - \frac{1}{2}gt^2, \qquad \textbf{Equation (4.4.4)}$$

$$v_y = v_0 \sin\theta_0 - gt \qquad \textbf{Equation (4.4.5)}$$

$$v_y^2 = (v_0 \sin\theta_0)^2 - 2g(y - y_0). \qquad \textbf{Equation (4.4.6)}$$

Projectile Trajectory: Definition

- The projectile's **trajectory** is
  - Its path through space (traces a parabola)
  - Found by eliminating time between Eqs. 4.4.3 and 4.4.4:

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0\cos\theta_0)^2}$$

Equation (4.4.7)

Where: θ_0 is the projectile angle, y is the vertical height at any time, x is the horizontal displacement at any time, and v_0 is the magnitude of the initial velocity.

Horizontal Range: Definition

- The **horizontal range** is:
  - The distance the projectile travels in *x* by the time it returns to its initial height

$$R = \frac{v_0^2}{g} \sin 2\theta_0.$$

Equation (4.4.8)

The horizontal range *R* is maximum for a launch angle of 45°.

Effect of Air Resistance

- In previous calculations we assumed air resistance is negligible, but in many situation that is a poor assumption.

Table 4.4.1 Two Fly Balls[a]

	Path I (Air)	Path II (Vacuum)
Range	98.5 m	177 m
Maximum height	53.0 m	76.8 m
Time of flight	6.6 s	7.9 s

[a]See Fig. 4.4.6. The launch angle is 60° and the launch speed is 44.7 m/s.

Effect of Air Resistance: Trajectories of Two Fly Balls

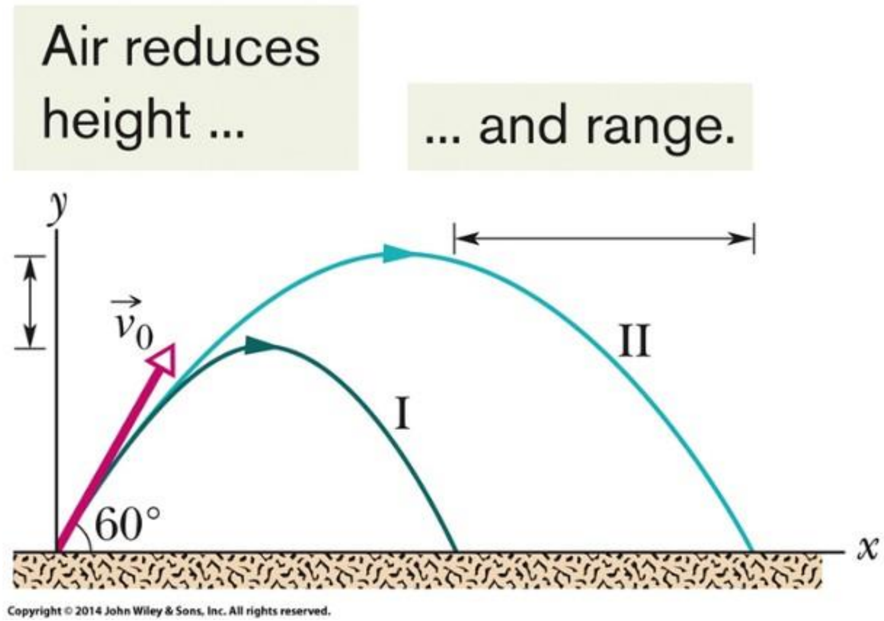


Figure 4.4.6

Projectile Motion Checkpoint #4

A fly ball is hit to the outfield. During its flight (ignore the effects of the air), what happens to its (a) horizontal and (b) vertical components of velocity? What are the (c) horizontal and (d) vertical components of its acceleration during ascent, during descent, and at the topmost point of its flight?

Answer:

(a) is unchanged

(b) decreases (becomes negative)

(c) 0 at all times

(d) $-g$ ($-$ 9.8 m/s^2) at all times

Section 4.5 Uniform Circular Motion

Uniform Circular Motion Learning Objectives

- Sketch the path taken in uniform circular motion and explain the velocity and acceleration vectors (magnitude and direction) during the motion.
- Apply the relationships between the radius of the circular path, the period, the particle's speed, and the particle’s acceleration magnitude.

Requirements for Uniform Circular Motion

- A particle is in **uniform circular motion** if
  - It travels around a circle or circular arc
  - At a constant speed
- Since the velocity changes, the particle is accelerating!
- Velocity and acceleration have:
  - Constant magnitude
  - Changing direction
- The constant change in direction of the velocity is the reason why acceleration is present, but speed is constant in the tangential direction.

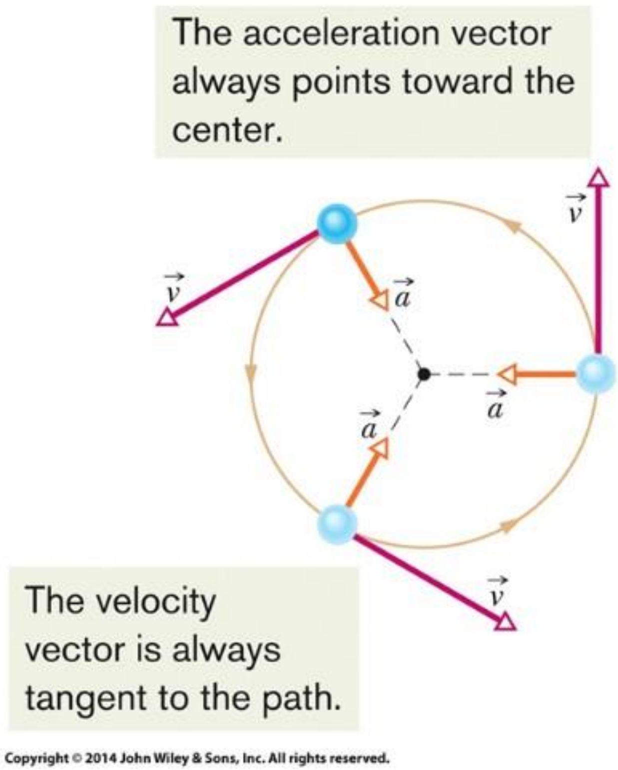


Figure 4.5.1

Centripetal Acceleration and Period of Revolution: Definitions

- Acceleration is called **centripetal acceleration**
  - Means "center seeking"
  - Directed radially inward

$$a = \frac{v^2}{r}$$ **Equation (4.5.1)**

- The **period of revolution** is:
  - The time it takes for the particle go around the closed path exactly once

where: T = period, r = radius of the circle, and v = velocity

$$T = \frac{2\pi r}{v}$$ **Equation (4.5.2)**

Uniform Circular Motion Checkpoint #5

An object moves at constant speed along a circular path in a horizontal *xy* plane, with the center at the origin. When the object is at $x = -2$ m, its velocity is $-\left(4\,{}^{m}\!/\!{}_{s}\right)\hat{j}$

Give the object's (a) velocity and (b) acceleration at $y = 2$ m.

Answer:

(a) $-\left(4\,{}^{m}\!/\!{}_{s}\right)\hat{i}$

(b) $-\left(8\,{}^{m}\!/\!{}_{s^2}\right)\hat{j}$

Section 4.6 Relative Motion in One Dimension

Relative Motion in One Dimension: Learning Objectives

- Apply the relationship between a particle's position, velocity, and acceleration as measured from two reference frames that move relative to each other at a constant velocity and along a single axis.

Role of Reference Frames in Determining Position and Velocity

- Measures of position and velocity depend on the **reference frame** of the measurer
  - How is the observer moving?
  - Our usual reference frame is that of the ground
- Read subscripts "*PA*", "*PB*", and "*BA*" as "*P* as measured by *A*", "*P* as measured by *B*", and "*B* as measured by *A*"
- Frames *A* and *B* are each watching the movement of object *P*

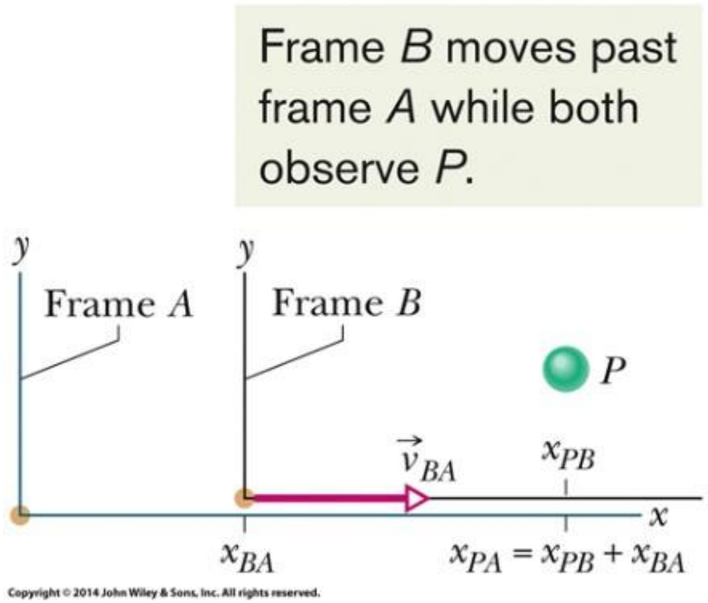


Figure 4.6.1

Position & Velocity in Two Different Reference Frames

- Positions in different frames are related by:

$$x_{PA} = x_{PB} + x_{BA} \qquad \textbf{Equation (4.6.1)}$$

- Taking the derivative, we see velocities are related by:

$$\frac{d}{dt}\left(x_{PA}\right) = \frac{d}{dt}\left(x_{PB}\right) + \frac{d}{dt}\left(x_{BA}\right) \qquad \textbf{Equation (4.6.2)}$$

$$v_{PA} = v_{PB} + v_{BA}$$

Acceleration with Respect to Two Reference Frames

- But accelerations (for non-accelerating reference frames, $a_{BA} = 0$) are related by

$$\frac{d}{dt}\left(v_{PA}\right) = \frac{d}{dt}\left(v_{PB}\right) + \frac{d}{dt}\left(v_{BA}\right)$$

$$a_{PA} = a_{PB}$$

Equation (4.6.3)

Relative Reference Frames: Particle Acceleration Measured by Frame A

Observers on different frames of reference that move at constant velocity relative to each other will measure the same acceleration for a moving particle.

Example

Frame A: $x = 2$ m, $v = 4$ m/s
Frame B: $x = 3$ m, $v = -2$ m/s

P as measured by A: $x_{PA} = 5$ m, $v_{PA} = 2$ m/s, $a = 1$ m/s^2

Relative Reference Frames: Particle Acceleration Measured by Frame B

So P as measured by B:

- $x_{PB} = x_{PA} + x_{AB} = 5\text{ m} + (2\text{m} - 3\text{m}) = 4\text{ m}$
- $v_{PB} = v_{PA} + v_{AB} = 2\text{ m/s} + (4\text{ m/s} - -2\text{m/s}) = 8\text{ m/s}$
- $a = 1\text{ m/s}^2$

Section 4.7 Relative Motion in Two Dimensions

Relative Motion in Two Dimensions Learning Objectives

- Apply the relationship between a particle's position, velocity, and acceleration as measured from two reference frames that move relative to each other at a constant velocity and in two dimensions.

Vector Representation of 2D Relative Motion

- The same as in one dimension, but now with vectors:
- Positions in different frames are related by:

$$\vec{r}_{PA} = \vec{r}_{PB} + \vec{r}_{BA}$$ **Equation (4.7.1)**

- Velocities:

$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$ **Equation (4.7.2)**

- Accelerations (for non-accelerating reference frames):

$$\vec{a}_{PA} = \vec{a}_{PB}$$ **Equation (4.7.3)**

- As was the case for 1D relative motion, observers in different frames will see the same acceleration

Graphical View of 2D Relative Motion

- Frames *A* and *B* are both observing the motion of *P*

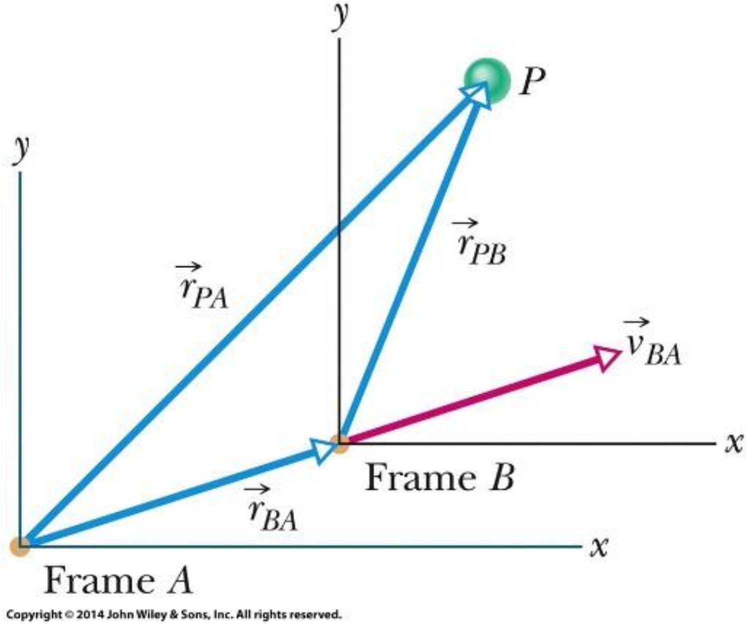


Figure 4.7.1

Chapter 4 Summary: Position & Displacement Vectors

Position Vector

- Locates a particle in 3-space

$$\vec{r} = x\hat{\mathrm{i}} + y\hat{\mathrm{j}} + z\hat{\mathrm{k}}$$ **Equation (4.1.1)**

Displacement

- Change in position vector

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$ **Equation (4.1.2)**

$$\Delta\vec{r} = (x_2 - x_1)\hat{\mathrm{i}} + (y_2 - y_1)\hat{\mathrm{j}} + (z_2 - z_1)\hat{\mathrm{k}}$$ **Equation (4.1.3)**

$$= \Delta x\hat{\mathrm{i}} + \Delta y\hat{\mathrm{j}} + \Delta z\hat{\mathrm{k}}$$

Equation (4.1.4)

Chapter 4 Summary: Average vs. Instantaneous Velocity and Acceleration

Average and Instantaneous Velocity

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{r}}{\Delta t}$$ **Equation (4.2.1)**

$$\vec{v} = \frac{d\,\vec{r}}{dt}$$ **Equation (4.2.3)**

Average and Instantaneous Accel.

$$\vec{a}_{\text{avg}} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t} = \frac{\Delta \vec{v}}{\Delta t}$$ **Equation (4.3.1)**

$$\vec{a} = \frac{d\vec{v}}{dt}.$$ **Equation (4.3.2)**

Chapter 4 Summary: Projectile Motion

Projectile Motion

- Flight of particle subject only to free-fall acceleration (g)

$$y - y_0 = v_{0y}t - \frac{1}{2}gt^2 \qquad \textbf{Equation (4.4.4)}$$

$$= (v_0 \sin\theta_0)t - \frac{1}{2}gt^2,$$

$$v_y = v_0 \sin\theta_0 - gt \qquad \textbf{Equation (4.4.5)}$$

- Trajectory is parabolic path

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0\cos\theta_0)^2} \qquad \textbf{Equation (4.4.7)}$$

Chapter 4 Summary: Projectile Horizontal Range

- Horizontal range:

$$R = \frac{v_0^2}{g} \sin 2\theta_0.$$

Equation (4.4.8)

Chapter 4 Summary: Uniform Circular Motion

Uniform Circular Motion

- Magnitude of acceleration:

$$a = \frac{v^2}{r}$$ **Equation (4.5.1)**

- Time to complete a circle:

$$T = \frac{2\pi r}{v}$$ **Equation (4.5.2)**

Chapter 4 Summary: Relative Motion

Relative Motion

- For non-accelerating reference frames

$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$ **Equation (4.7.2)**

$$\vec{a}_{PA} = \vec{a}_{PB}$$ **Equation (4.7.3)**

Biomedical Applications: Problem 24

In the 1991 World Track and Field Championships in Tokyo, Mike Powell jumped 8.95 m, breaking by a full 5 cm the 23-year long-jump record set by Bob Beamon. Assume that Powell's speed on takeoff was 9.5 m/s (about equal to that of a sprinter) and that g = 9.80 m/s2 in Tokyo. How much less was Powell's range than the maximum possible range for a particle launched at the same speed?

Answer: 0.259 m

Biomedical Applications: Problem 51

A skilled skier knows to jump upwards before reaching a downward slope. Consider a jump in which the launch speed is v_0 = 10 m/s, the launch angle is θ_0 = 11.3°, the initial course is approximately flat, and the steeper track has a slope of 9.0°. Figure 4.22 (a) below shows a *prejump* that allows the skier to land on the top portion of the steeper track. Figure 4.22 (b) below shows a jump at the edge of the steeper track. In Figure 4.22 (a), the skier lands at approximately the launch level.

Question (a) In the landing, what is the angle ϕ between the skier's path and the slope? Question (b) How far below the launch level does the skier land. Question (c) what is the angle ϕ (The greater fall and greater ϕ can result in low of control in the landing.)

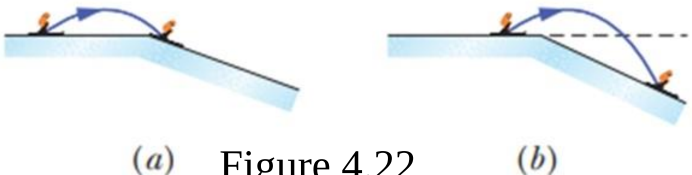


Figure 4.22

Answer: (a) 2.3°, (b) 1.1 m, (c) 18°

Biomedical Applications: Problem 71

A suspicious-looking man runs as fast as he can along a moving sidewalk from one end to the other, taking 2.50 s. Then security agents appear, and the man runs as fast as he can back along the sidewalk to his starting point, taking 10.0 s. What is the ratio of the man's running speed to the sidewalk's speed?

Answer: 1.67